

Computer Science - Undergraduate Level Assessment 1 Total Marks: 40 Answer all questions.

Long Form Response (40 marks)

Provide thorough reasoning and reference key steps.

1. State the basic laws of Boolean algebra.
2. Statement 1— The Stanford Sentiment Treebank contained movie reviews, not book reviews. Statement 2— The Penn Treebank has been used for language modeling.
3. Let $x = 1$. What is $x \&\& 3$ in Python 3?
4. Consider $x(t)$ to be given as, $x(t) = 10 \cos(20\pi t - \pi/4) - 5 \cos(50\pi t)$ What is minimum sampling rate (/Hz) such that $y(t) = x(t)$?

End of Paper